IS883: Deploying Generative AI

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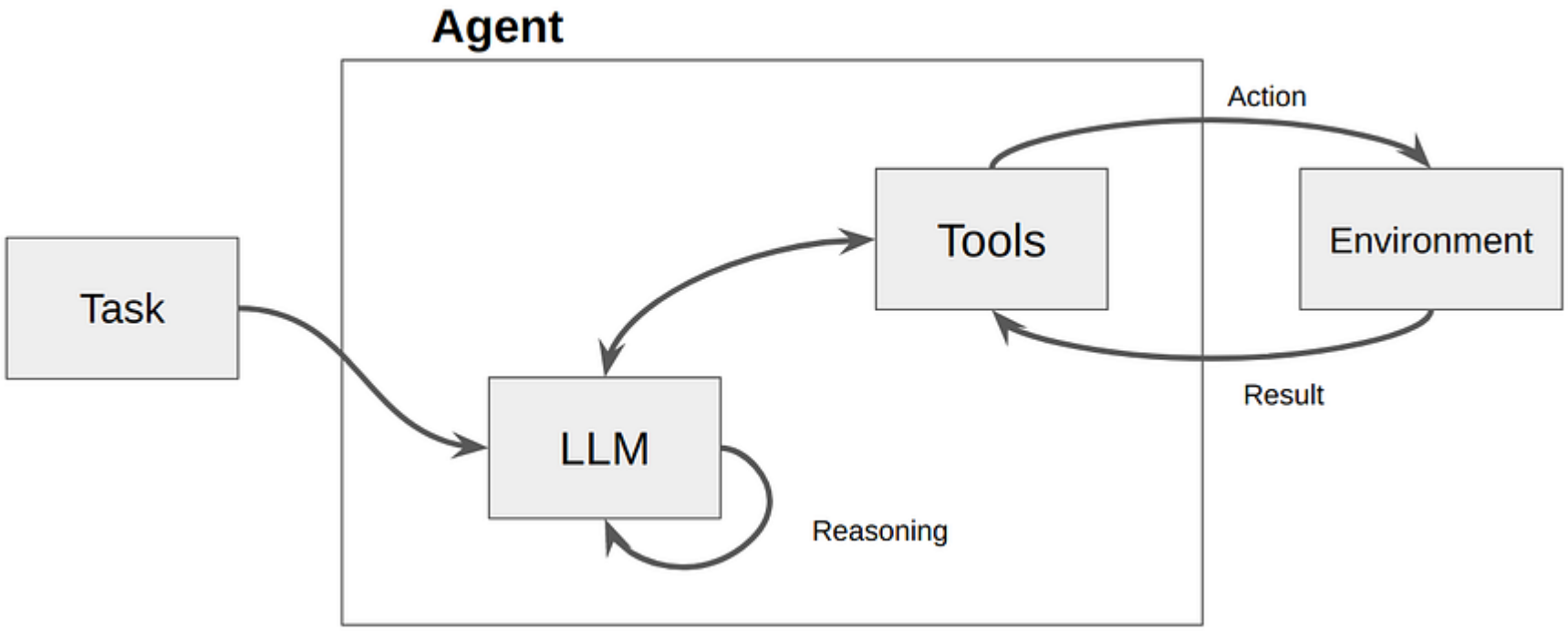
What is an Agent?

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Demo

- Try some of the following prompts:
  - “Who is <*some obscure relative of yours*>*?*”
  - Calculate the product of the following numbers <*an array of large numbers*>

How Agents Work?



betterprogramming.pub/

What are Tools?

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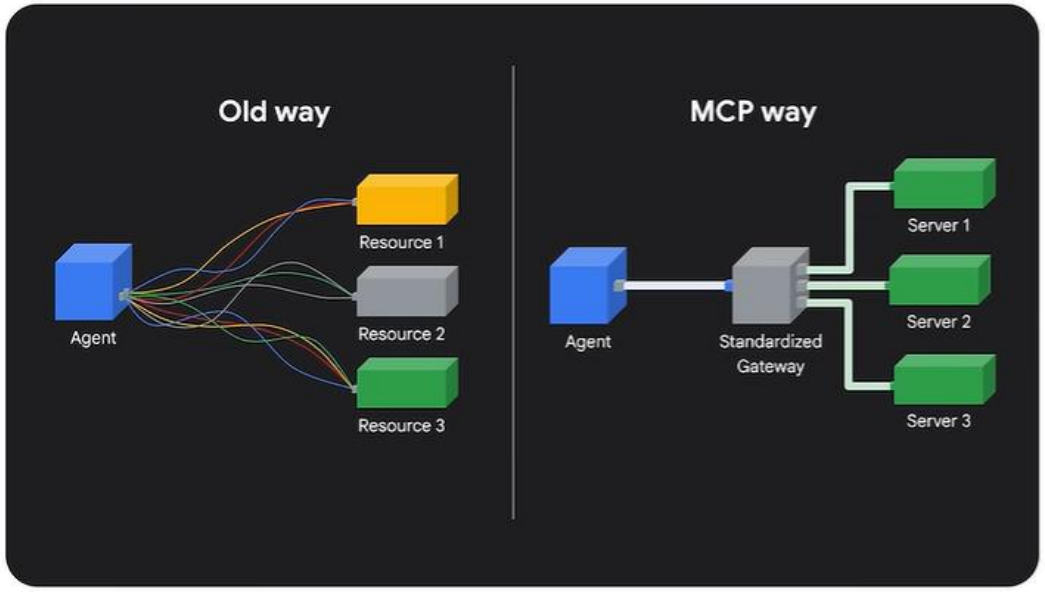
How Agents Work

- Examples of some tools.
- The model needs to "parse" the input text into arguments that can be used by the tool (e.g., API)



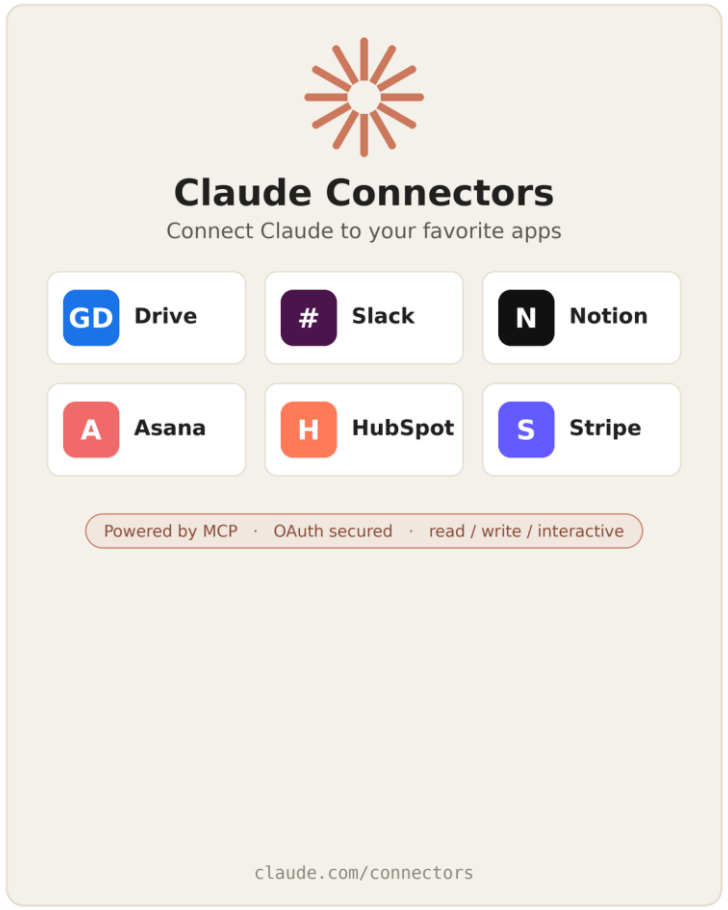
Tools in Practice: MCP

- MCP (Model Context Protocol): an open standard from Anthropic for connecting AI to external tools and data.
- mcp.so is a community directory of ready-made MCP servers (terminals, databases, agent memory, healthcare, and more).
  - Some provide MCP servers to use, while others provide open-source code to self-host.



Tools in Practice: Connectors

- Connectors = MCP made one-click, built right into Claude.
- Connect Claude to your favorite apps s, such as Google Drive, Slack, Notion, Stripe, etc.
- Claude can then read (and optionally write) your apps during a conversation



Tools in Practice: Skills

- A Skill = packaged expertise (instructions + a repeatable workflow) an agent can reuse. It is an advanced form of prompt engineering.
- Skill marketplaces let you install community skills into Claude in one click. (e.g., skillsmp.com. Take a look at finance-billing-ops)

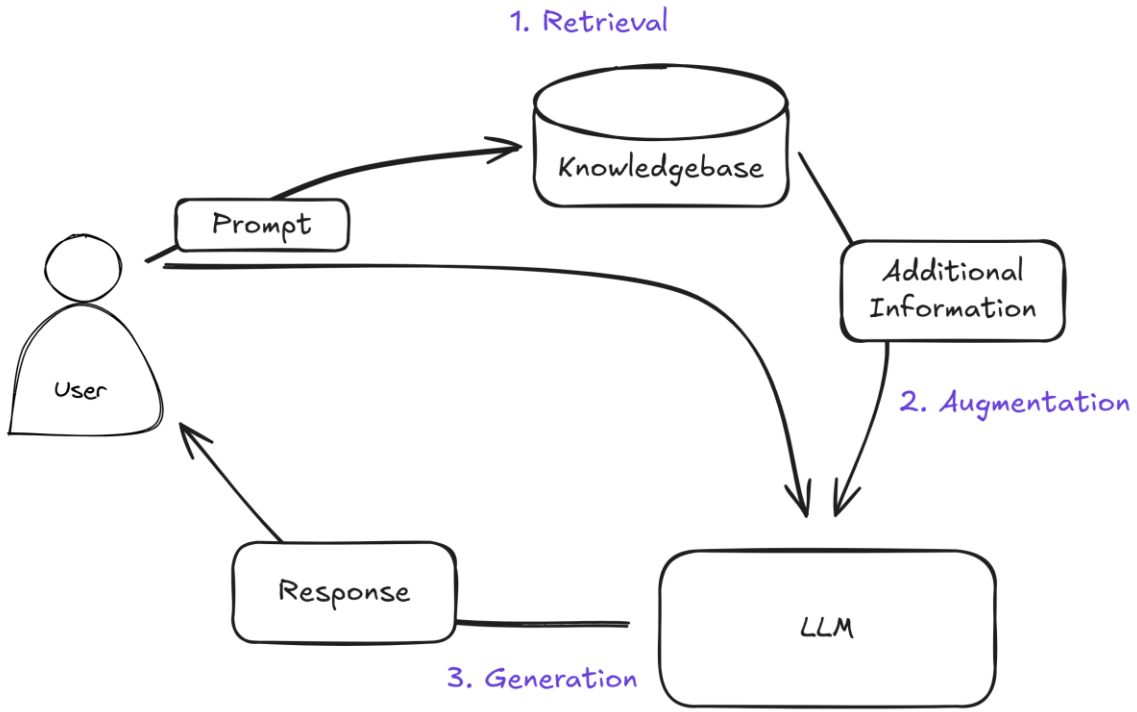
How to Access External knowledge?

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Incorporating Knowledge

- In its most basic form, the LLM only completes the prompt based on its internal parameters.
- Tools can provide more context for the LLM to make a better completion.
- More/better context could be provided through knowledge(e.g., text in files)



NVIDIA's Last Quarter

Prompt: "What was NVIDIA's total revenue last quarter?" — no internet, no search tool.

Without RAG

Answers from its parameters alone — the newest quarter that was in its training data.

$46.7B

Q2 FY2026 · quarter ended July 27, 2025

Fluent, confident, and off by $34.9B.

With RAG

Retrieves the paragraph from NVIDIA's earnings release and answers from that text.

$81.6B

Q1 FY2027 · quarter ended April 26, 2026 · up 85% YoY

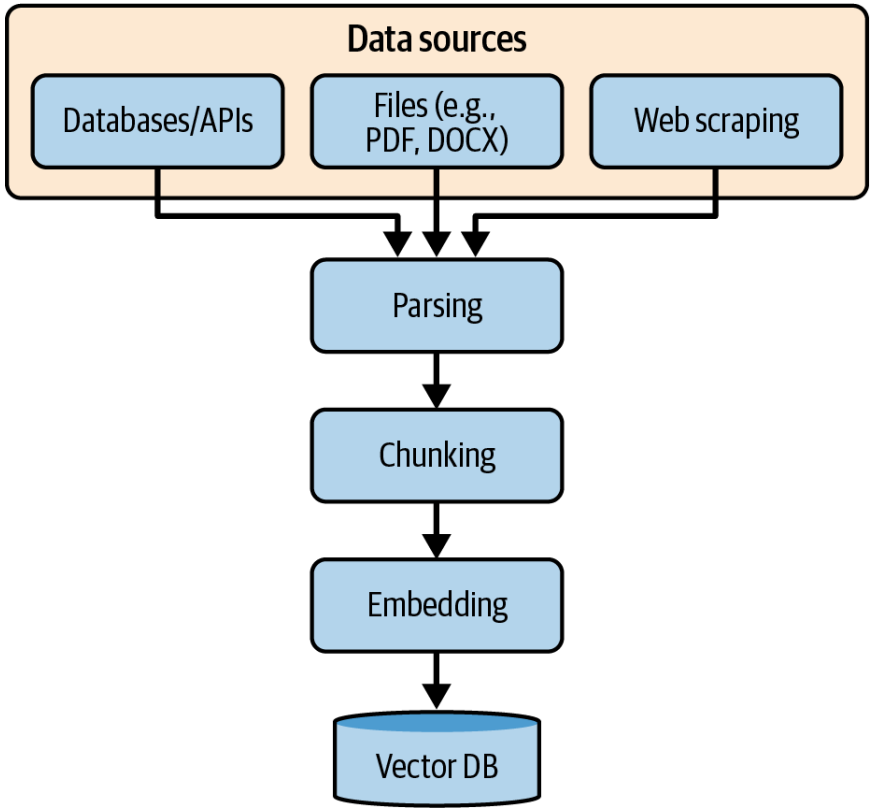
Correct — and you can check the source.

Same model, same question. The only thing that changed is what was in the prompt.

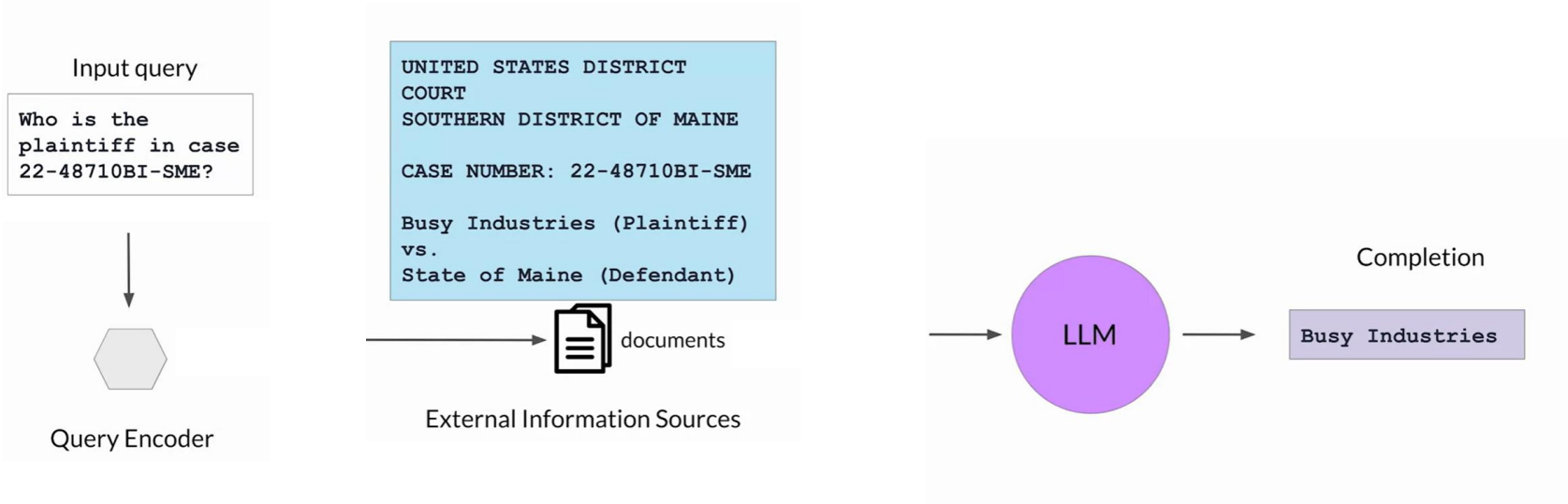
Source: NVIDIA, "NVIDIA Announces Financial Results for First Quarter Fiscal 2027," nvidianews.nvidia.com. Retrieved figure: revenue of $81,615 million, up 85% from a year ago.

Knowledge is Overwhelming!

- Knowledge may contain huge amounts of data, most of which may not be relevant.
- The model needs to find the *chunks* that are relevant to the prompt in question.
- Finding the appropriate chunk(s):
  - Storing all chunks as embeddings in a Vector database
  - Extracting the chunks with high similarity to the prompt (as measured by cosine similarity)
- This is called **RAG**

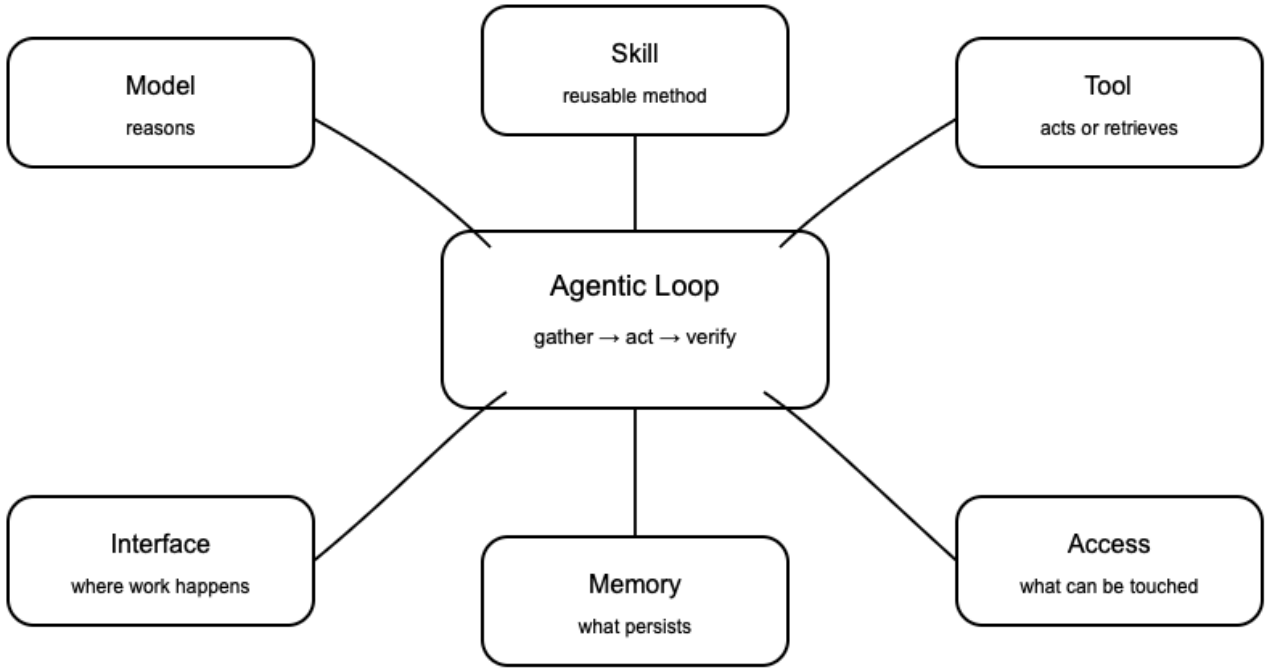


Example



coursera.org

The Big Picture



The model does not work alone. The surrounding layers shape what agentic work can do.